ST 4: Destillasjon

TKP 4105 - Separasjonsteknikk
11. september 2024

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1 Introduction

In this experiment, distillation using Raschig rings will be conducted on a mixture containing 10% mol ethanol and water. The goal of the experiment is to determine the time required for the column to reach steady-state conditions, and to evaluate the column's efficency as a function of vapour velocity. The purpose of the experiment is to provide insights to the fundamental principles of distillation, to understand how to achieve optimal operating conditions, and understand the key parameters influencing the operation of a distillation column.

2 Theory

2.1 Destillation

Destillation is a method used to purify or separate liquids from eachother. [1] The separation is based on the components having different boiling points, the mixed liquids is heated up so one of the components vaporizes and is separated from the liquid.

A packed column is a type of vertical column used for distillation and other separation processes. It contains one or more sections of packing material that increases the surface area for contact between phases, such as liquid and vapor. Fluid flows downwards and vapor rises upwards through the column, this promotes repeated vapor-liquid contact. [2]

2.1.1 Reflux

Reflux is a process used to to improve the separation of components in a mixture. It involves returning a portion of the cooled down vapor to the top of the distillation column to flow back down through the packing or trays. This helps the temperature of the reaction stay constant. This experiment operates with total reflux, witch means that all vapor is returned to the column. [3]

2.1.2 Column efficiency

The column efficiency in a distillation column is defined as the ratio of the number of ideal equilibrium stages (theoretical plates) to the number of actual trays in the column. If the outgoing liquid and vapor phases reach equilibrium on each tray, then the distillation process is considered ideal. High column efficiency implies good contact between the gas and liquid phases, resulting in effective separation.

Since the separation process in a packed column does not occur in discrete trays or stages, the Height Equivalent to a Theoretical Plate (HETP), shown i Likning (2.1), is used to estimate the equivalent number of theoretical stages (or "actual trays"). The HETP represents the height of packing material that achieves the same separation as one theoretical plate in a tray column. [4]

$$\text{HEPT} = \frac{\text{packed height}}{\text{number of equivalent equilibrium stages}} \quad [2] \tag{2.1}$$

2.1.3 Flooding point

Flooding is defined as a state where the liquid no longer can flow downward due to an overload of vapour going up. This results in the liquid being carried out of the column with the vapor or the liquid building up in the column. Therefore there is a maximum vapor flow rate for every given liquid flow rate, this is the flooding point. [2]

2.1.4 McCabe-Thiele method

The McCabe-Thiele method is a graphic method used to determine the number of quilibrium stages, reflux ratio, and optimal feed stage location in a binary destillation. It's based on the

constants molar overflow assumption, that implies constant molar vapor and liquid flow rate leaving all stages in a section, and that the pressure is isobaric. It combines the equilibrium curve with operating lines to design and analyze distillation columns and helps visualize the distillation process and calculate necessary column specifications. [4]

2.2 Gas Chromatography

Gas chromatography is a method used to separate and analyze organic compounds. A injected gas phase is transported by a inert carrier gas (mobile phase) trough a column with a stationary phase. [5] The stationary phase is a liquid with a high boiling point or a solid adsorbent. As the sample moves through the column, different compounds interact differently with the stationary phase and thus separate based on their volatility and affinity for the stationary phase. After separation, the compounds exit the column and enter a detector. [6]

3 Experimental

The destilationns lab was a two part lab. The first part of the lab was to determine how long the column takes to reach steady state condition, and the second part was to determine the efficiency of the column.

3.1 Column starting

- Turn on the PC and the turn on the cooling water
- Make sure that the bottom and distillate tap lines are closed.
- Fill the column with 1596.1 mL ethanol and 4203.9 mL water, and set the column to total reflux.
- Set the temperature to 105 °C.
- When the temperature changes substantially, or the pressure difference over the column rises quickly, set the boilers' power to 50%.
- Start the time when the vapour stream starts condensing.

In case of flooding conditions, switch off the heater immediately. In any emergency situation, turn off the heater but leave the cooling water on.

3.2 Exercise nr. 1

- Take samples from the top after 2, 6 and 10 minutes, and then every 5 minutes within 1 hour, totaling in 13 top-samples.
- After the 13th sample take a bottom sample to analyze in the next exercise. Note down the temperature and pump flow rate at the extraction time.
- Analyse all samples and find out how long the column takes to reach steady state

3.3 Exercise nr. 2

- Set the reboiler to 40% power
- Wait until the column reaches the steady-state. This will be at the same time as in exercise 1.
- Take samples from the top and from the bottom, 3-4 ml each. Write down the reflux rate at the time of sampling.
- Repeat two more times with different boiler power settings.

- Analyse the samples and calculate the vapour velocity, the efficiency of the column, and the theoretical number of stages using McCabe-Thiele.

3.4 Column Shutdown

- Turn off the heater.
- Let the column with contents cool down.
- Empty the column and turn off the cooling water.
- Turn off the rest of the equipment.

3.5 Gas Chromatography Analysis

- Prepare the samples for the GC by mixing 500 μL of ethanol sample with 250 μL of isopropanol in a glass vial.
- Seal the vials properly with a cap.
- Analyse the samples using gas chromatography

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A Calculations

A.1 A mixture of H_2O and EtOH

The physical properties needed when calculating the volumetric composition of the mix of water and ethanol, are shown in table A.1. Mm is molar mass, ρ is the density, and x is the component's mole fraction.

Tabell A.1: Physical properties for H_2O and EtOH

Component	$Mm[gmol^{-1}]$	$\rho[g/cm^3]$	x
EtOH	46.040	0.789	0.1
H_2O	18.016	0.997	0.9

The mixture's total volume, $V_{tot} = 5800mL$, can be expressed as:

$$V_{tot} = V_{\mathrm{H_2O}} + V_{EtOH}, \tag{A.1}$$

where $V_{\mathrm{H_2O}}$ is the volume of H_2O, and V_{EtOH} is the volume of the ethanol. The mole fraction for ethanol in the mixture is given by:

$$x_{EtOH} = \frac{V_{\mathrm{H_2O}}}{n_{EtOH} + n_{\mathrm{H_2O}}}, \tag{A.2}$$

where n is the number of moles of the respective components. The equations A.3 and A.4 are used to calculate the moles for water and ethanol respectively.

$$n_{\mathrm{H_2O}} = \frac{V_{\mathrm{H_2O}} \cdot \rho_{\mathrm{H_2O}}}{Mm_{\mathrm{H_2O}}}, \tag{A.3}$$

$$n_{\mathrm{EtOH}} = \frac{V_{\mathrm{EtOH}} \cdot \rho_{\mathrm{EtOH}}}{Mm_{EtOH}}, \tag{A.4}$$

where V is the components volume. Putting the equations A.1, A.3, and A.4 into A.2, to get the following equation:

$$X_{EtOH} = \frac{\frac{V_{\mathrm{EtOH}} \cdot \rho_{\mathrm{EtOH}}}{Mm_{EtOH}}}{\frac{V_{\mathrm{EtOH}} \cdot \rho_{\mathrm{EtOH}}}{Mm_{EtOH}} + \frac{(V_{tot} - V_{EtOH}) \cdot \rho_{\mathrm{H_2O}}}{Mm_{\mathrm{H_2O}}}} \tag{A.5}$$

By inserting the variables from table A.1, A.5 can be solved for V_{EtOH}, which gives:

$$V_{EtOH,100\%} = 1532.3mL \tag{A.6}$$

Since 96% ethanol is used in the experiment, the calculated value above needs to be divided by 0.96 in order to find the correct volume,

$$V_{EtOH,96\%} = 1596.1mL \tag{A.7}$$

With $V_{EtOH,96\%}$ known,. it's possible to find $V_{\mathrm{H_2O}}$ by using the equation A.1,

$$V_{\mathrm{H_2O}} = 4203.9mL \tag{A.8}$$

B Example calculations

B.1 Bottom compositions and distillate from pycnometers

It is desired to calculate the mole fractions of ethanol in both the distillate and bottom product, by using pycnometers, and comparing with the GC calibrations results. To find the equation needed, A.5 can be rewritten as the following:

$$\begin{aligned} x_{EtOH} &= \frac{\frac{m_{EtOH}}{Mm_{EtOH}}}{\frac{m_{EtOH}}{Mm_{EtOH}} + \frac{m_{H_2O}}{Mm_{H_2O}}} \\ &= \frac{\frac{wt\%_{EtOH}}{Mm_{EtOH}}}{\frac{wt\%_{EtOH}}{Mm_{EtOH}} + \frac{wt\%_{H_2O}}{Mm_{H_2O}}}, \end{aligned} \tag{B.1}$$

where $wt\%$ is the weight percentage of the different components. In order to calculate x_{EtOH}, $wt\%_{EtOH}$ must first be found. To calculate the densities of the distillate and bottom product, the equations B.2 and B.3, respectively, are used.

$$\rho_{distillate} = \frac{\rho_{water} \cdot m_{distillate}}{m_{water}} = \frac{(16.0903 - 11.98170) \cdot 0.997}{16.95955 - 11.98170} = 0.8229 g/cm^3 \tag{B.2}$$

$$\rho_{bottom} = \frac{\rho_{water} \cdot m_{bottom}}{m_{water}} = \frac{(16.0903 - 1.77385) \cdot 0.997}{16.95955 - 1.77385} = 0.9847 g/cm^3 \tag{B.3}$$

The ethanol-water density-composition data at 25°C [7], can be used to estimate $wt\%_{EtOH}$ in the distillate and the bottom product, based on the calculated densities. For the distillate, the weight percentage is estimated to be 88%, and for the bottom product, the weight percentage is estimated tp be 8%. These values can be put into B.1 in order to calculate the mole fractions of ethanol.

$$X_{EtOH,distillate} = 0.741$$

$$X_{EtOH,bottom} = 0.033$$

These values are similar to the ones from the GC analysis, which means that the two methods lead to approximately the same results.

B.2 Vapour Velocity

The formula for calculating the vapour velocity is given by:

$$v = \frac{\dot{V}_{gas}}{A}, \tag{B.4}$$

where $\dot{V}_{gas}$ is the volume rate of the gas, and A is the cross-sectional area of the Column. To calculate A, the following equation is used:

$$A = \frac{\pi}{4} D^2, \tag{B.5}$$

where D is the column's diameter. The ideal gas law can be used to calculate the volume rate, $\dot{V}_{gas}$,

$$\dot{V}_{gas} = \frac{\dot{n}RT}{p}, \tag{B.6}$$

where $\dot{n}$ is the molar gas flow, R is the universal gas constant, T is the temperature, and p is the pressure. The vapour velocity can be calculated by putting the equations B.5 and B.6 into B.4:

$$v = \frac{4\dot{n}RT}{\pi D^2 p}, \tag{B.7}$$

the diameter of the column, D is given as $0.05m$, the temperature is 25℃, and the pressure is $1Bar$. In order to calculate v, the molar flow needs to be found

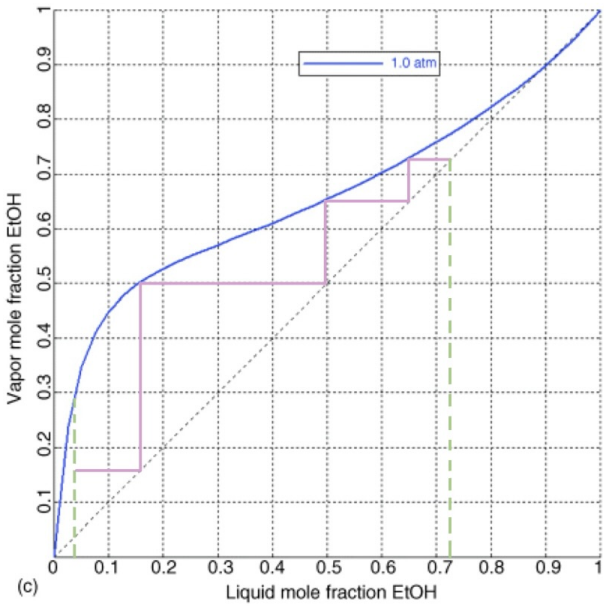
$$\begin{aligned} \dot{n} = \frac{\dot{m}}{Mm} &= \frac{\dot{V} \cdot \rho_{distillate}}{X_{EtOH} \cdot Mm_{EtOH} + (1 - X_{EtOH}) \cdot Mm_{\mathrm{H_2O}}} \\ &= \frac{53 \cdot 0.8229}{0.741 \cdot 46.07 + 0.259 \cdot 18.016} = 1.124 molmin^{-1} = 0.0187 mols^{-1} \end{aligned} \tag{B.8}$$

From this, the vapour velocity can be calculated by inserting the given and calculated values into B.7:

$$v = \frac{4 \cdot 0.0187 \cdot 8.314 \cdot 298}{\pi \cdot 0.05^2 \cdot 10^5} = 0.24 ms^{-1} \tag{B.9}$$

B.3 Number of ideal stages using McCabe-Thiele

Figure B.1 shows the McCabe-Thiele diagram for water and ethanol. Both the lower and upper operating line are equal to $y = x$, given total reflux. The equilibrium-line was found using data from Vapor-Liquid Equilibrium Data Collection [8]. The number of ideal stages is approximately 3.8.



Figur B.1: McCabe-Thiele diagram for a mixture of water and ethanol.

C HSE

The chemicals and gasses used in this lab are Ethanol, isopropanol, helium, hydrogen gas and compressed air. Ethanol and isopropanol both are highly flammable, can cause eye irritation and dizziness. The use of personal protective equipment such as the use of protective gloves when handling the chemicals, along with a lab coat, goggles and closed toe shoes, is recommended.

Other potential risks include high temperatures and chipped or broken glass. Gas detectors will alert in case of H_2-gas leakage, and the lab assistant should be notified. If equipment of high temperatures is not handled responsibly, it may lead to burn injuries. Glass equipment may break and cause small cuts.